Crash course — Payload CMS + Astro for QLU News

A short primer so you know **what** each piece does, **how** they work together, and **why** this stack fits Quantum Life University's magazine + Jnanodayam site.

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1. Headless in one sentence

Payload stores and edits content (admin + API). **Astro** draws the public website by reading that API. The browser never runs WordPress PHP — only the pages Astro builds or serves.

```
Editors  →  Payload Admin (/admin)
                 ↓
            PostgreSQL + media files
                 ↓
            REST API (/api/…)
                 ↓
Readers  →  Astro site (flipbook magazine, Jnanodayam pages)
```

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2. Payload CMS (the brain)

What it is: An open-source, TypeScript **headless CMS** that runs on Node (this repo: Next.js app in `apps/cms`). You define **collections** (like custom post types); Payload gives you an admin UI and a REST/GraphQL API for free.

What we use it for:

Collection	Role
`issues`	Monthly magazine (`YYYY-MM`), Te/En titles, sections, images
`jnanodayam-readings`	Series + day readings (e.g. Devi Navaratri)
`media`	Photos / uploads
`users`	Editor logins

Things to remember:

- **Admin** = `http://localhost:3000/admin` — Lexical rich text, image align/size, publish checkbox.
- **API** = `http://localhost:3000/api/issues`, `/api/jnanodayam-readings`, etc.
- Content lives in **Postgres**, not in git. Folder packs under `content/` are for **seed/import**, then you edit in admin.
- Only **published** docs are public; logged-in editors can see drafts.

Mental model: Payload ≈ “WordPress admin + database,” without forcing the public site to be WordPress themes/plugins.

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3. Astro (the face)

What it is: A **web framework** focused on content sites. By default it ships little or no JavaScript to the browser. Pages are `.astro` files (HTML-like + frontmatter).

What we use it for:

Route	Experience
`/`	Home — latest issue + Jnanodayam entry
`/issues/…`	Magazine archive + **landscape flipbook** reader
`/jnanodayam/…`	Programme hub + day readings Te/En

SSR (server-side render): Magazine/Jnanodayam dynamic URLs fetch Payload **on each request**, so a publish in admin shows up on refresh — no full static rebuild required for v1.

Mental model: Astro ≈ “the designed reader” — soft-paper look, flipbook, Te|En — while Payload holds the raw content.

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4. Why this pair fits QLU News

Need	How Payload + Astro help
Monthly magazine with fixed sections	Structured fields (`masthead` … `closing`), not freeform page soup
Telugu + English	Parallel Te/En fields on the same issue/reading
Jnanodayam series	Separate collection (series + day), same soft-paper brand
Editors without HTML	Lexical rich text + image layout in admin
Nice reading experience	Astro owns layout (flipbook, chips, landscape) — CMS doesn’t dictate theme hacks
Smaller public attack surface	No PHP plugin zoo on the public site; harden Node + Postgres instead
Seed from folders	Same pack idea as the WordPress repo; `npm run seed` fills Payload
Host on one VPS	CMS + Astro + Postgres via Docker on Hostinger KVM (see hosting doc)

Compared to classic WordPress: WP is still great for drag-and-drop blocks. This stack wins when you want **typed content**, a **custom reader UI**, and a clearer split between "edit" and "display."

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5. Day-to-day flow

1. **Edit** in Payload admin (or re-seed from `content/magazine/YYYY-MM/`).
2. **Publish** the issue / reading.
3. **Open** the Astro site — flipbook or Jnanodayam page loads fresh data from the API.

Local: `.\dev.cmd` → Postgres + Payload `:3000` + Astro `:4321`.

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6. Glossary (tiny)

Term	Meaning here
Collection	A content type (issues, readings, media)
Lexical	Payload's rich-text editor (stores JSON, we render HTML on the site)
SSR	Server renders the page when someone visits (live CMS data)
Seed	Script that imports folder packs into Payload
Headless	CMS has no mandatory public theme; any front-end can consume the API

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7. Where to go next

Doc	Topic
LOCAL.md	Install, ports, seed
CONTENT.md	Folder packs, admin fields, URLs
ARCHITECTURE.md	Apps and data flow
Hosting	Hostinger VPS
CMS-ALTERNATIVE.md	Why headless vs WordPress (sibling repo)

Official deep dives: payloadcms.com/docs · docs.astro.build